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Q5) Prove part 2 & 3 of Theorem 4.1.3.

2) $P \in A \times (B \cup C)$

$\rightarrow \exists x \exists y [x \in A \wedge y \in B \cup C \wedge p = (x, y)]$

$x_1 \in A$

$y_1 \in B \vee y_1 \in C$

$p = (x_1, y_1)$

$\therefore$ existential
inst

$P \in (A \times B) \cup (A \times C)$

$P \in A \times B \vee P \in A \times C$

~~$x \in A$~~

$(x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$

Case 1) Supp $y_1 \in B$ then $x_1 \in A$ so $p = (x_1, y_1) \in A \times B$

Case 2) Supp $y_1 \in C$ then $x_1 \in A$ so $p \in A \times C$

Supp $P \in (A \times B) \cup (A \times C)$

$P \in A \times (B \cup C)$

$(x \in A \wedge y \in B) \vee (x \in A \wedge y \in C)$

$x \in A \wedge (y \in B \vee y \in C)$

Case 1) $x \in A \wedge y \in B$ so $y \in B \cup C$ since $y \in B$. So $P \in A \times (B \cup C)$

Case 2) $x \in A \wedge y \in C$ similar

3) Supp $P \in (A \times B) \cap (C \times D)$

$P \in (A \cap C) \times (B \cap D)$

$\rightarrow P \in A \times B \wedge P \in C \times D$

$x \in A \cap C \wedge y \in B \cap D$

$(x \in A \wedge y \in B) \wedge (x \in C \wedge y \in D)$

$x \in A \wedge x \in C \wedge y \in B \wedge y \in D$

easy to follow

← Same 1

1) $\{(x, y) \in P \times D \mid x \text{ is parent of } y\} = \{(Adnan, Umar), (Mona, Umar), \dots\}$

1b) $\{(x, y) \in T \times U \mid \text{There is someone who divides in } x \text{ \& attends } y\} = \{(university, PUCIT), \dots\}$

Mergestack

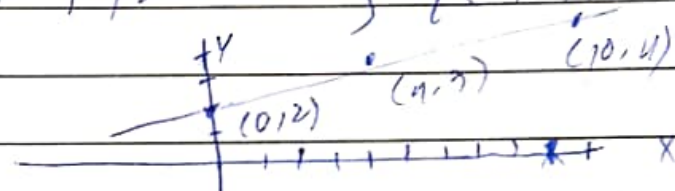
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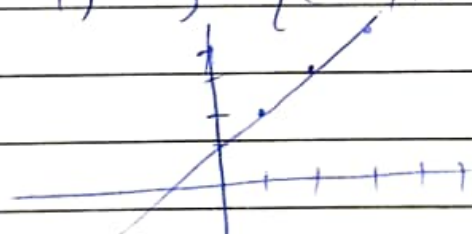
① 2 (a) $\{(x, y) \in \mathbb{R}^2 \mid x \text{ lives in } y\}$
 or the person x lives in city y
 $= \{(man, lahore) \dots\}$

(b) $\{(x, y) \in (x, N) \mid \text{the city } x \text{ has } y \text{ population}\}$
 $= \{(lahore, 4000000) \dots\}$

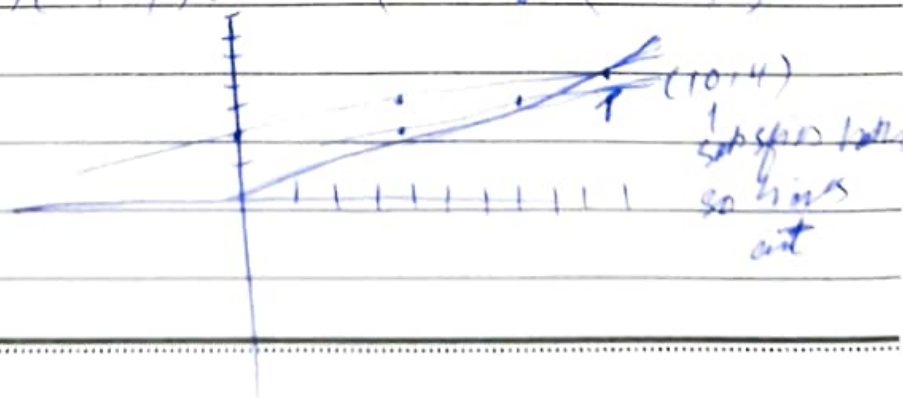
3) $\{(x, y) \in \mathbb{R}^2 \mid y = x^2 - x - 2\} = \{(0, 2), (4, 3), (10, 4) \dots\}$



(b) $\{(x, y) \in \mathbb{R}^2 \mid y < x\} = \{(1, 2), (2, 3), (3, 4) \dots\}$



(c) $\{(x, y) \in \mathbb{R}^2 \mid y = x^2 - x - 2 \vee y = 3x - 2\}$
 $= \{(0, 2), (4, 2), (4, 3), (10, 4), (7, 3)\}$

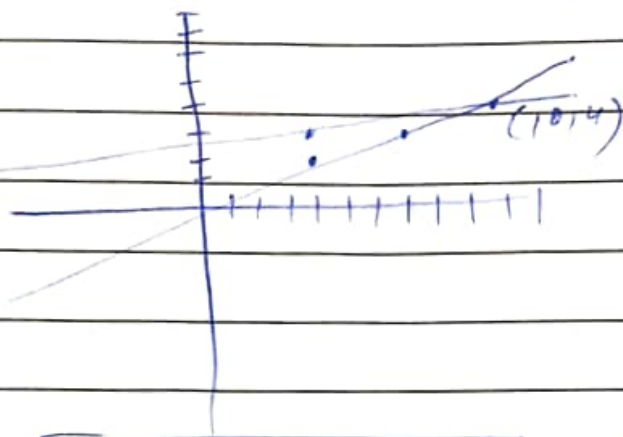


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$$(d) \{(x, y) \in \mathbb{R}^2 \mid y \in \mathbb{N} \wedge (y = x^2 - x - 2 \vee y = 3x - 2)\}$$

$$= \{(4, 2), (4, 3), (7, 3), (10, 4), \dots\}$$



$$4) A = \{1, 2, 3\} \quad B = \{1, 4\} \quad C = \{3, 4\} \quad D = \{5\}$$

$$a) A \times B \cap C = \{1, 2, 3\} \times \{4\}$$

$$= \{(1, 4), (2, 4), (3, 4)\}$$

$$(A \times B) \cap (A \times C) = \{(1, 4), (2, 4), (3, 4), (1, 3), (2, 3), (3, 3)\}$$

$$\cap \{(1, 4), (2, 4), (3, 4)\}$$

$$= \{(1, 4), (2, 4), (3, 4)\}$$

$$(d) (A \times B) \cup (C \times D) = \{(1, 1), (2, 1), (3, 1), (1, 4), (2, 4), (3, 4)\} \cup \{(3, 5), (4, 5)\}$$

$$= \{\text{all of these}\}$$

$$A \cup C = A \cup B \cap D$$

$$(A \cup C) \times (B \cup D) = \{1, 2, 3, 4\} \times \{1, 4, 5\}$$

$$= \{(1, 1), (1, 4), (1, 5), (2, 1), \dots, (4, 4), (4, 5)\}$$

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6)

$$1) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

→ Supp $P \in A \times (B \cap C)$

$$x \in A \wedge y \in B \wedge y \in C \quad \text{--- (1)}$$

We have to prove that

$$P \in (A \times B) \cap (A \times C) \vee [P \in (A \times B) \wedge P \notin (A \times C)]$$

$$(x \in A \wedge y \in B) \wedge (x \in A \vee y \in C) \vee (x \in A \wedge y \in C) \wedge (x \notin A \vee y \notin C)$$

by (1) This follows

$$\leftarrow (A \times B) \cap (A \times C) \rightarrow A \times (B \cap C)$$

$$x \in A \wedge y \in B \wedge y \in C$$

Case 1) $x \in A \wedge y \in B \wedge (x \notin A \vee y \in C)$ Since $x \in A$ so $y \in C$. Goal follows.

Case 2) $x \in A \wedge y \in C \wedge (x \notin A \vee x \in B)$. Since $x \in A$ so $x \in B$. Goal doesn't follow.

$$\rightarrow x \in A \wedge y \in B \wedge y \in C$$

$$x \in A \wedge y \in B \wedge \neg(x \in A \wedge y \in C)$$

goal follows

$$x \in A \wedge y \in B \wedge (x \notin A \vee y \notin C)$$

$$\leftarrow x \in A \wedge y \in B$$

$$x \in A \wedge y \in B \wedge y \notin C$$

$$x \notin A \vee y \notin C$$

Case 3) as $x \in A$ so $y \notin C$. So goal follows.

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$$9) A \times (B \cap C) = (A \times B) \cap (A \times C)$$

$x \in A \wedge y \in B \cap C$
 $(x, y) \in (A \times B) \cap (A \times C)$

$$(x \in A \wedge y \in B) \wedge (x \in A \wedge y \in C)$$

$$[(x, y) \in (A \times B) \wedge (x, y) \in (A \times C)] \vee [(x, y) \in (A \times C) \wedge (x, y) \in (A \times B)]$$

$$(x \in A \wedge y \in B \wedge (x \in A \vee y \in C))$$

$$\vee (x \in A \wedge y \in C \wedge (x \in A \vee y \in B))$$

Case 1) $y \in B \wedge y \in C$. Goal part ① follows.

Case 2) $y \notin B \wedge y \in C$. part ② follows.

← Case 1) Supp $x \in A \wedge y \in B \wedge (x \notin A \vee y \notin C)$. Then since $x \in A$ so $y \notin C$. Goal follows as $x \in A \wedge y \in B \wedge y \notin C$.

Case 2) Supp $x \in A \wedge y \in C \wedge (x \notin A \vee y \notin B)$. Since $x \in A$ so $y \notin B$. Goal follows.

$$10) (A \setminus C) \times (B \setminus D) \subseteq (A \times B) \setminus (C \times D)$$

$$\forall p [p \in \text{" } \rightarrow p \in \text{"}]$$

$$\text{Supp } p \in (A \setminus C) \times (B \setminus D)$$

$$x \in A \setminus C \wedge y \in B \setminus D$$

$$x \in A \wedge x \notin C \wedge y \in B \wedge y \notin D$$

$$p \in (A \times B) \wedge p \notin (C \times D)$$

$$x \in A \wedge y \in B \wedge (x \notin C \vee y \notin D)$$

follows.

11)

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$$11) (A \times B) \setminus (C \times D) = [A \times (B \setminus D)] \cup [(A \setminus C) \times B]$$

→

$$x \in A \wedge y \in B \wedge (x \notin C \vee y \notin D) \quad \text{--- ①}$$

we have to prove that

$$p \in A \setminus (B \setminus D) \vee p \in (A \setminus C) \times B$$

$$(x \in A \wedge y \in B \wedge y \notin D) \vee (x \notin A \wedge x \notin C \wedge y \in B)$$

From ①

Case 1) $x \notin C$ part ② of goal follows.

Case 2) $y \notin D$ part ① of goal follows

← Supp

$$(x \in A \wedge y \in B \wedge y \notin D) \vee (x \in A \wedge x \notin C \wedge y \in B)$$

$$\begin{cases} x \in A \\ y \in B \\ x \notin C \vee y \notin D \end{cases}$$

Case 1) Goal follows

Case 2) Goal follows.

$$12) \forall p [p \in A \times B \rightarrow p \notin C \times D] \quad \forall y [y \in A \rightarrow y \notin C]$$

$$\forall (x, y) [x \in A \wedge y \in B \rightarrow (x \notin C \vee y \notin D)] \vee \forall y [y \in B \rightarrow y \notin D]$$

$$\text{Supp } x \in A, x \in B$$

$$(x \notin C) \vee$$

$$x \notin C \vee x \notin D \text{ for } \forall x$$

$$(x \notin D)$$

as arbitrary so goal follows

To understand this type of goal better, consider the formal
 union. Let x be arbitrary element of A . Let z be arbitrary
 element of B . Then $(x, z) \in (A \times B)$ so $(z, z) \notin C \times D$

which means either $z \notin C$ or $z \notin D$ so either $A \setminus C$ or $B \setminus D$ are disjoint

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13) $I \neq \emptyset$

$$(\cap_{i \in I} A_i) \times B = \cap_{i \in I} (A_i \times B)$$

$$\rightarrow x \in \cap_{i \in I} A_i \wedge y \in B$$

$$\cap_{i \in I} (x \in A_i \wedge y \in B)$$

$$\forall i \in I (x \in A_i) \quad \textcircled{1}$$

$$\forall i \{ i \in I \rightarrow x \in A_i \wedge y \in B \}$$

$$y \in B \quad \textcircled{2}$$

$$\text{Supp } i \in I$$

$$x \in A_i \wedge y \in B$$

As $I \neq \emptyset$ $\exists i \in I$ so $x \in A_i$ by $\textcircled{1}$

I also will have $y \in B$ by $\textcircled{2}$

←

$$\text{Supp } \forall i \{ i \in I \rightarrow (x \in A_i \wedge y \in B) \} \quad \textcircled{1}$$

$$\forall i \{ i \in I \rightarrow x \in A_i \}$$

$$\wedge y \in B$$

$$\text{Suppose } i \in I$$

$$x \in A_i \wedge y \in B$$

as $I \neq \emptyset$ so $\textcircled{1}$ is not false

so $x \in A_i \wedge y \in B$ by $\textcircled{1}$

$$14) (a) \cup_{i \in I} (A_i \times B_i) \subseteq (\cup_{i \in I} A_i) \times (\cup_{i \in I} B_i)$$

$$\rightarrow \cup_{i \in I} (x \in A_i \wedge y \in B_i)$$

$$x \in \cup_{i \in I} A_i \wedge y \in \cup_{i \in I} B_i$$

$$\exists i \{ i \in I \wedge x \in A_i \wedge y \in B_i \}$$

$$\exists i \in I (x \in A_i) \wedge \exists i \in I (y \in B_i)$$

By existential inst, goal follows.

←

$$\text{Supp } \exists i \in I (x \in A_i) \wedge \exists i \in I (y \in B_i)$$

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(b) ~~$\forall (i,j) \in I \times I$~~ $(i,j) = A_i \times B_j$

$P = I \times I \rightarrow$

$x_0 \in I \wedge y_0 \in I \wedge p = (x, y)$

We have to prove that

$U_{p \in P} C_p = (U_{i \in I} A_i) \times (U_{j \in I} B_j)$

Supp $q = (x, y) \in U_{p \in P} C_p$

$q \in U_{i \in I} A_i \times U_{j \in I} B_j$

$\exists p \in P (q \in C_p)$

$x \in U_{i \in I} A_i \wedge y \in U_{j \in I} B_j$

By universal inst

$\exists i (i \in I \wedge x \in A_i) \wedge \exists j (j \in I \wedge y \in B_j)$

$p_0 \in P \wedge q \in C_{p_0}$

As $P = I \times I$

so $p_0 \in I \times I$ which means $x_0 \in I$ & $y_0 \in I$

$q \in C_{p_0}$ means $q \in C_{x_0 y_0}$ which means $q \in A_{x_0} \times B_{y_0}$ as

$(i, j) = A_i \times B_j$ - so $q \in A_{x_0} \times B_{y_0}$ as $q = (x, y)$

so we had $x_0 \in I$ & $x_0 \in A_{x_0}$ & $y_0 \in I$ & $y_0 \in B_{y_0}$

Let $q = (x, y)$

$\leftarrow$ Supp $\exists i \in I (x \in A_i)$

$\exists j \in I (y \in B_j)$

$u \in I \wedge x \in A_u$ - ①

$v \in I \wedge y \in B_v$ - ②

Goal follows

$q \in U_{p \in P} C_p$

$\exists p \in P (q \in C_p)$

$\exists p \in (I \times I) (q \in C_p)$

$\exists i \in I \wedge \exists j \in I \wedge q \in C_{i \times j}$

" $q \in A_i \times B_j$

" $x \in A_i \wedge y \in B_j$

if we used universal here
will be used here as ~~universal~~
Quantifier is on whole
thing